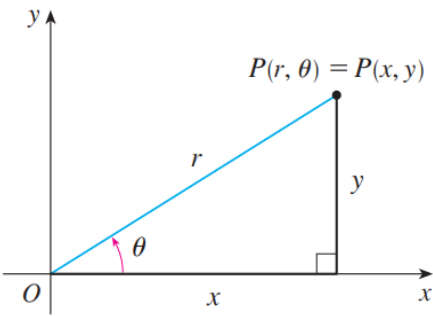


15.4 Double Integrals in Polar Coordinates

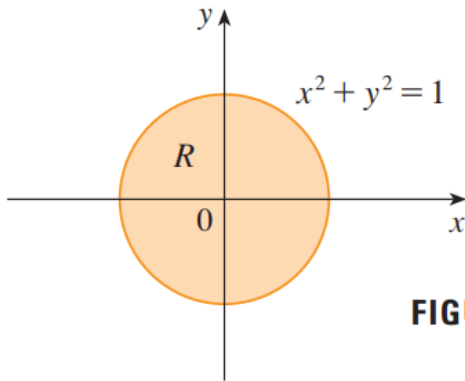


$$r^2 = x^2 + y^2 \quad x = r \cos \theta \quad y = r \sin \theta$$

The regions in the next **Figure** are special cases of a **polar rectangle**

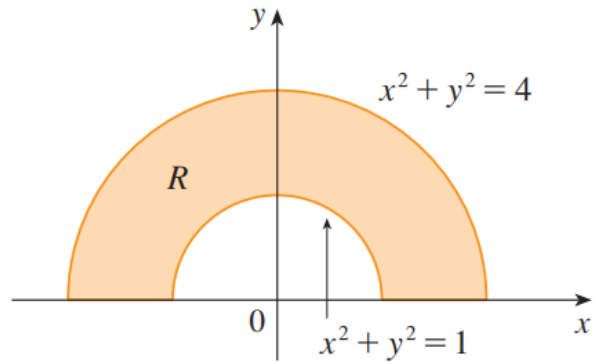
$$R = \{(r, \theta) \mid a \leq r \leq b, \alpha \leq \theta \leq \beta\}$$

(Constant limits for double integration) which is easy to integrate – that is why we will use it



(a) $R = \{(r, \theta) \mid 0 \leq r \leq 1, 0 \leq \theta \leq 2\pi\}$

FIGURE



(b) $R = \{(r, \theta) \mid 1 \leq r \leq 2, 0 \leq \theta \leq \pi\}$

15.5 Applications of Double Integrals

probability density function f of a continuous random variable X is $f(x)$ such that

$$f(x) \geq 0 \text{ for all } x, \int_{-\infty}^{\infty} f(x) dx = 1,$$

and the probability that X lies between a and b is found by integrating f from a to b

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

Now we consider a pair of continuous random variables X and Y , such as the lifetimes of two components of a machine or the height and weight of an adult female chosen at random. The **joint density function** of X and Y is a function f of two variables such that the probability that (X, Y) lies in a region D is

$$P((X, Y) \in D) = \iint_D f(x, y) dA$$

In particular, if the region is a rectangle, the probability that X lies between a and b and Y lies between c and d is

$$P(a \leq X \leq b, c \leq Y \leq d) = \int_a^b \int_c^d f(x, y) dy dx$$

Because probabilities aren't negative and are measured on a scale from 0 to 1, the joint density function has the following properties:

$$f(x, y) \geq 0 \quad \iint_{\mathbb{R}^2} f(x, y) dA = 1$$

Suppose X is a random variable with probability density function $f_1(x)$ and Y is a random variable with density function $f_2(y)$. Then X and Y are called **independent random variables** if their joint density function is the product of their individual density functions:

$$f(x, y) = f_1(x) f_2(y)$$

Usually, waiting times are modeled by using exponential density functions

$$f(t) = \begin{cases} 0 & \text{if } t < 0 \\ \mu^{-1} e^{-t/\mu} & \text{if } t \geq 0 \end{cases}$$

where μ is the mean waiting time.

EXAMPLE The manager of a movie theater determines that the average time moviegoers wait in line to buy a ticket for this week's film is 10 minutes and the average time they wait to buy popcorn is 5 minutes. Assuming that the waiting times are independent, find the probability that a moviegoer waits a total of less than 20 minutes before taking his or her seat.

waiting time X for the ticket purchase and the waiting
time Y in the refreshment line

Since X and Y are independent, the joint density function is the product:

$$f(x, y) = f_1(x) f_2(y) = \begin{cases} \frac{1}{50} e^{-x/10} e^{-y/5} & \text{if } x \geq 0, y \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

We are asked for the probability that $X + Y < 20$:

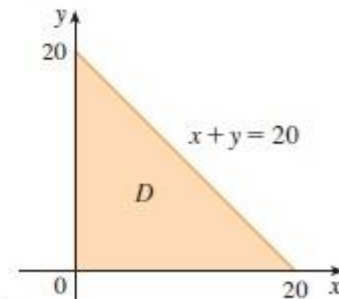
$$P(X + Y < 20) = P((X, Y) \in D)$$

where D is the triangular region shown in Figure Thus

$$P(X + Y < 20) = \iint_D f(x, y) dA = \int_0^{20} \int_0^{20-x} \frac{1}{50} e^{-x/10} e^{-y/5} dy dx$$

$$= 1 + e^{-4} - 2e^{-2} \approx 0.7476$$

This means that about 75% of the moviegoers wait less than 20 minutes before taking their seats.



FIGURE

Expected Values

if X is a random variable with probability density function f , then its *mean* is

$$\mu = \int_{-\infty}^{\infty} xf(x) dx$$

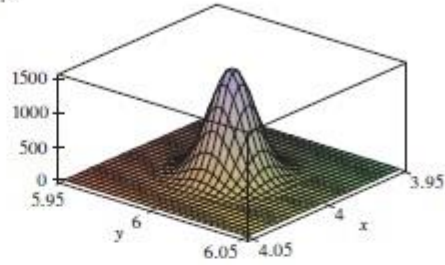
Now if X and Y are random variables with joint density function f , we define the **X -mean** and **Y -mean**, also called the **expected values** of X and Y , to be

$$\mu_1 = \iint_{\mathbb{R}^2} xf(x, y) dA \quad \mu_2 = \iint_{\mathbb{R}^2} yf(x, y) dA$$

a single random variable is *normally distributed* if its probability density function is of the form

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}$$

where μ is the mean and σ is the standard deviation.



Graph of the bivariate normal joint density function

EXAMPLE A factory produces (cylindrically shaped) roller bearings that are sold as having diameter 4.0 cm and length 6.0 cm. In fact, the diameters X are normally distributed with mean 4.0 cm and standard deviation 0.01 cm while the lengths Y are normally distributed with mean 6.0 cm and standard deviation 0.01 cm. Assuming that X and Y are independent, write the joint density function and graph it. Find the probability that a bearing randomly chosen from the production line has either length or diameter that differs from the mean by more than 0.02 cm.

SOLUTION We are given that X and Y are normally distributed with $\mu_1 = 4.0$, $\mu_2 = 6.0$, and $\sigma_1 = \sigma_2 = 0.01$. So the individual density functions for X and Y are

$$f_1(x) = \frac{1}{0.01\sqrt{2\pi}} e^{-(x-4)^2/0.0002} \quad f_2(y) = \frac{1}{0.01\sqrt{2\pi}} e^{-(y-6)^2/0.0002}$$

Since X and Y are independent, the joint density function is the product:

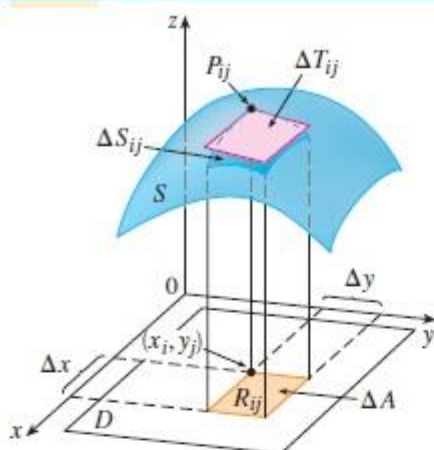
$$\begin{aligned} f(x, y) &= f_1(x)f_2(y) \\ &= \frac{1}{0.0002\pi} e^{-(x-4)^2/0.0002} e^{-(y-6)^2/0.0002} = \frac{5000}{\pi} e^{-5000[(x-4)^2+(y-6)^2]} \end{aligned}$$

Let's first calculate the probability that both X and Y differ from their means by less than 0.02 cm. Using a calculator or computer to estimate the integral, we have

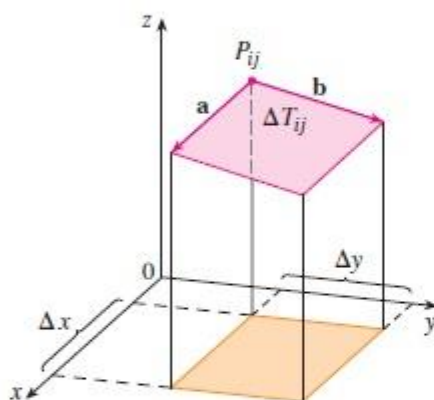
$$\begin{aligned} P(3.98 < X < 4.02, 5.98 < Y < 6.02) &= \int_{3.98}^{4.02} \int_{5.98}^{6.02} f(x, y) dy dx \\ &= \frac{5000}{\pi} \int_{3.98}^{4.02} \int_{5.98}^{6.02} e^{-5000[(x-4)^2+(y-6)^2]} dy dx \\ &\approx 0.91 \end{aligned}$$

Then the probability that either X or Y differs from its mean by more than 0.02 cm is approximately $1 - 0.91 = 0.09$

15.6 Surface Area



FIGURE



FIGURE

$$A(S) = \lim_{m, n \rightarrow \infty} \sum_{i=1}^m \sum_{j=1}^n \Delta T_{ij}$$

$$\mathbf{a} = \Delta x \mathbf{i} + f_x(x_i, y_j) \Delta x \mathbf{k}$$

$$\mathbf{b} = \Delta y \mathbf{j} + f_y(x_i, y_j) \Delta y \mathbf{k}$$

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \Delta x & 0 & f_x(x_i, y_j) \Delta x \\ 0 & \Delta y & f_y(x_i, y_j) \Delta y \end{vmatrix}$$

$$= -f_x(x_i, y_j) \Delta x \Delta y \mathbf{i} - f_y(x_i, y_j) \Delta x \Delta y \mathbf{j} + \Delta x \Delta y \mathbf{k}$$

$$= [-f_x(x_i, y_j) \mathbf{i} - f_y(x_i, y_j) \mathbf{j} + \mathbf{k}] \Delta A$$

$$\Delta T_{ij} = |\mathbf{a} \times \mathbf{b}| = \sqrt{[f_x(x_i, y_j)]^2 + [f_y(x_i, y_j)]^2 + 1} \Delta A$$

$$A(S) = \lim_{m, n \rightarrow \infty} \sum_{i=1}^m \sum_{j=1}^n \Delta T_{ij}$$

$$= \lim_{m, n \rightarrow \infty} \sum_{i=1}^m \sum_{j=1}^n \sqrt{[f_x(x_i, y_j)]^2 + [f_y(x_i, y_j)]^2 + 1} \Delta A$$

The area of the surface with equation $z = f(x, y)$, $(x, y) \in D$, where f_x and f_y are continuous, is

$$A(S) = \iint_D \sqrt{[f_x(x, y)]^2 + [f_y(x, y)]^2 + 1} dA$$

$$A(s) = \iint_D \sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2} dA$$

EXAMPLE Find the area of the part of the paraboloid $z = x^2 + y^2$ that lies under the plane $z = 9$.

Converting to polar coordinates, we obtain

$$\begin{aligned} A &= \int_0^{2\pi} \int_0^3 \sqrt{1 + 4r^2} r dr d\theta \\ &= \frac{\pi}{6} (37\sqrt{37} - 1) \end{aligned}$$